

Varun Bharadwaj

Mechanical & Aerospace Design Engineer

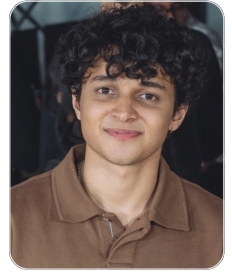
Structures • Mechanisms • Thermomechanical FEA • Materials

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Profile

Mechanical and aerospace design engineer who likes working from first principles and taking hardware all the way from requirements through CAD and simulation to real, tested parts. Most of my work has been in precision mechanisms, lightweight composite and metallic structures, thermal protection, and the structural FEA that goes with them: sizing parts for static and dynamic loading, vibration and shock, and thermomechanical behaviour at high temperature. Experience across aerial robotics, electric aircraft and prosthetics, with a habit of picking up unfamiliar systems quickly and turning analysis into parts that actually get built.

Experience

Apr 2026
– Present
Munich, DE

Stealth Startup, Mechanical Engineer

Aerial robotics company. Heavyweight firefighting UAV (250 kg class).

- **Primary mechanical designer on a 250 kg firefighting UAV:** architecture and structural layout from conceptual through detailed design, translation of commercial requirements into engineering ones, and overall development and integration of the aircraft and its next version.
- Designed a **two-axis gimballed nozzle** from a clean sheet that aims and fires the high-pressure water jet in flight, covering the kinematics, swivel joints, bearing arrangements and load paths.
- **Sized and specified its actuation:** worked out the torque budget against the jet reaction and slew requirements, selected the servos, designed the couplings and mounts, then programmed and calibrated them on the aircraft.
- Sized the airframe and tank interfaces around **water sloshing**, where the coupled fluid and structural loads drive stiffness and fatigue, and traced a flight instability back to slosh dynamics rather than controller tuning.
- Design **field-deployable, weather-resistant hardware:** IP67 electronic enclosures and IP54 structural assemblies, with sealing, gaskets, drainage and materials and finishes suited to outdoor operation.
- Work with a **product designer** on packaging, ergonomics and serviceability, so the aircraft stays easy to handle and maintain in the field without losing function.
- Designed the **battery integration**, pack mounting and high-current harness routing.
- Design, simulate and tolerance parts, then release them for **CNC machining, sheet metal and 3D printing**, iterating on flight and field feedback with the electronics team. Maintain the **DMU** and run the design in **3DEXPERIENCE** for configuration control and PLM.
- **Fly the aircraft and run the flight test campaigns;** EU A2 and **STS** remote-pilot certificates for multirotor operations.

Jul 2024
– Apr 2026
Munich, DE

Vaeridion GmbH, Mechanical Design Engineer

Propulsion battery system for a CS-23 all-electric aircraft.

- Designed **structural battery pack components and integrated hardware** (support structures, mounting interfaces, enclosure elements), from concept through prototype to validation.
- Co-developed a **patent-pending sandwich-structure concept for interstitial cell support** that combines structural load-carrying capability with strong thermal runaway performance. Named as a **co-inventor on the filed patent application**.
- Ran **structural and thermomechanical FEA** in Abaqus, explicit and implicit, for stiffness, load paths and robustness, including **temperature-dependent material models** above **1000 °C**.
- Built a reduced stiffness and mass model of the cell stack to run **global eigenmode analyses**, keeping the module's modes clear of airframe excitation for **vibration qualification** (UN38.3).
- Developed **thermal runaway mitigation concepts:** sealing, venting, containment and structural protection features for aerospace battery systems.
- Planned and ran the **structural and thermal runaway test campaigns:** designed the test articles, rigs, fixtures and manufacturing jigs, and prototyped quickly with 3D printing and laser cutting.
- Designed for **manufacture and assembly** and drove fabrication with suppliers and the internal workshop across mechanical, electrical, thermal and test teams. Automated parameter sweeps and post-processing in Python and MATLAB.

May 2024 – Oct 2024 Munich, DE	MIRMI (TUM), Mechanical Design Engineer Robotic prosthetic hand for Cybathlon. - Worked on the mechanical design of a robotic prosthetic hand, taking mechanisms from concept to working prototype in a very tight package. - Designed the finger joints and transmission: pivots, linkages, bearing and bushing choices, and drive routing through the hand, keeping friction and backlash down. - Sized and integrated the actuation, matching motors and gearing to the grip force and speed needed, and packaged them with the wiring inside the hand. - Ran grip and actuation tests on the bench, measured how the mechanism actually behaved against what we designed for, and fed that into the next revision. - Iterated SLA and FDM prototypes quickly, working through tolerance-driven fit and function issues with a multidisciplinary team.
Aug 2023 – Jan 2025 Munich, DE	HORYZN (TUM), Aerostructures & CAD Team Lead Student initiative building eVTOL and UAV aircraft. 20-person team. Led the aerostructures and CAD work on two aircraft, the GoAero emergency-response eVTOL and the “Kolibri” AED-delivery UAV, from concept through detailed design to manufactured, flying assemblies. Led a 20-person team: split and coordinated the work, set the CAD standards, and chaired the preliminary and critical design reviews. - Designed the lightweight composite airframe structures, structural layout, joint concepts and assembly approach; planned layups and supported manufacturing. - Secured supplier partnerships and funding and handled sponsor-facing technical communication. - Ran prototype planning and integration troubleshooting with the flight and avionics teams, then flew the aircraft on test.
Apr 2024 – Sep 2024 Munich, DE	TUM Autonomous Aerial Systems, Flight-Test & Integration Engineer (SPRIND Funke) Designed a payload gripping mechanism for a hexacopter and supported avionics and system integration, running pre-flight checks and quick fixes during field tests.
Mar 2023 – Aug 2023 Munich, DE	HORYZN (TUM), Software & Avionics Engineer Implemented autonomy features (path planning and precision landing) in ROS and Python , and supported avionics integration and flight software testing.
Oct 2022 – Feb 2023 Munich, DE	WARR (TUM), Structural Engineer Composite manufacturing and structural prototyping for a bipropellant liquid rocket, including regeneratively cooled chamber subsystems and layout and tooling planning.
Nov 2023 – Feb 2024 Munich, DE	TUM, Engineering Mechanics Tutor Tutored undergraduates in statics, dynamics and strength of materials, and put together reusable solution templates for exam preparation.

Education

Apr 2026 – Present Munich, DE	M.Sc. Aerospace Engineering, Technische Universität München (TUM) Modules: Analysis & Design of Composites · Aeroelasticity · Flight Dynamics, Stability & Control · Orbital Robotic Systems · Rotorcraft · Rocket Propulsion · Re-entry Aerodynamics.
Oct 2022 – Feb 2026 Munich, DE	B.Sc. Aerospace Engineering, Technische Universität München (TUM) Coursework: Structures & Optimization · Engineering Mechanics · Flight Mechanics & Control · Aircraft Design · Aerodynamics · Propulsion · Thermodynamics & Heat Transfer.
2021 – 2022 Goa, IN	B.E. (Year 1), Electronics & Communication Eng., BITS Pilani, Goa

Technical Skills

Design & Hardware	Concept-to-detail design · DFM / DFA · GD&T & tolerance stack-ups · Packaging & ergonomics · DMU
Mechanisms & Actuation	Joints, linkages & kinematics · Servo & motor sizing · Gearing & backlash · Bearings & swivels · Torque budgets
Structures & Materials	Composites & layup · Adhesive bonding · Lightweight structures · Thermal protection · Material selection
FEA & Simulation	Abaqus (explicit & implicit) · HyperMesh / OptiStruct · ANSYS · SolidWorks Simulation · Static, modal, dynamic & thermal
Rugged Hardware	Environmental sealing · IP-rated enclosures (IP67 / IP54) · Gaskets & venting · Weather-resistant materials & finishes
Manufacturing	CNC machining · Sheet metal · SLA & FDM 3D printing · Laser cutting · Jigs & tooling · Supplier coordination
Test & Validation	Bench & actuation testing · Test rigs & fixtures · Field & flight test · Failure investigation · EU A2 & STS remote pilot
CAD & Programming	CATIA V5 · Siemens NX · SolidWorks · 3DEXPERIENCE (CAD / PLM) · Python · MATLAB / Simulink · ROS
Languages & Awards	English, Kannada Native · Hindi Advanced · German Intermediate · NTSE Scholar · JEE Advanced, Top 1% (India)